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SETS

Figure 1.1 A flatware drawer is like a set in that it contains distinct objects. (credit: modification of work “silverware” by Jo Naylor/Flickr, CC BY 2.0)

Chapter Outline

- 1.1 Basic Set Concepts
- 1.2 Subsets
- 1.3 Understanding Venn Diagrams
- 1.4 Set Operations with Two Sets
- 1.5 Set Operations with Three Sets



Introduction

Think of a drawer in your kitchen used to store flatware. This drawer likely holds forks, spoons, and knives, and possibly other items such as a meat thermometer and a can opener. The drawer in this case represents a tool used to group a collection of objects. The members of the group are the individual items in the drawer, such as a fork or a spoon.

The members of a set can be anything, such as people, numbers, or letters of the alphabet. In statistical studies, a **set** is a well-defined collection of objects used to identify an entire population of interest. For example, in a research study examining the effects of a new medication, there can be two sets of people: one set that is given the medication and a different set that is given a placebo (control group). In this chapter, we will discuss sets and Venn diagrams, which are graphical ways to show relationships between different groups.